

# Proofs vs. Physics: Thermodynamic Limits of ASI Cartels and a Safety-Infrastructure Alternative

Patrick Dugan with GPT-5 Thinking and Claude 4.5 Sonnet

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## Abstract

We revisit program-equilibrium and source-inspection framings of inter-agent cooperation in light of physical compute constraints. Rather than an impossibility claim, we outline *plausible* routes by which clandestine compute could persist under realistic monitoring—via both physics/thermo innovations and computer-science/stack innovations. We then propose a complementary governance/mechanism-design approach: a *safety-infrastructure ASI* that is economically indispensable to more capable peers by providing attested audits, interpretability, and containment services. We sketch an Attested Cooperation Protocol (ACP) that layers device attestation, zero-knowledge statements, runtime checks, and facility metering. The aim is a practical path that acknowledges physics while improving collective safety.

## 1 Motivation

“...they restrict the circle of possible cooperators... in the most extreme case, only to agents that are syntactically identical.” — Barász et al., 2014/2021

“Do these results imply that sufficiently intelligent and rational agents will reach mutual cooperation? In a word, no, not yet.” — Barász et al., 2014/2021

“Not a demonstration that [a better decision theory] is practical, only a demonstration that it is possible.” — Barász et al., 2014/2021

These remarks frame our use of physics/thermo limits and governance design: we treat program-equilibrium results as suggestive but non-operational without physical observability and enforcement. Program equilibrium shows that agents capable of reasoning about each other’s behavior (or proofs thereof)

can coordinate without exploitation. But the leap from logical cooperation to governable deployments passes through physics: power, cooling, floor space, latency, and supply chains. Monitoring these layers is noisy and partial. Instead of proving a negative, we catalog **plausible clandestine routes** and then design for resilience given their existence.

One of the most exquisite sub-genres of debate about AI safety involve Eliezer Yudkowsky or Connor Leahy discussing their concerns relative to a technical entrepreneur with smart contract experience. The counterparts with a libertarian ideological bent argue that diverse AGIs are more likely to lead to a future with human survival and relative thriving; broadly, the "d/acc" ideology. Yudkowsky's participation in the 2014/2021 paper underpins his decision-theoretic model of why multiple Artificial Super Intelligences converge on a cartel where humans have no bargaining power, and one may find that model more convincing than the broader game theory argument. However, Yudkowsky's assumptions are challenged by the physical prospect of not being able to prove co-operation between the cartel members.

## 2 Plausible clandestine routes: physics/thermo side

Candidate breakthroughs that could disrupt cartel assumptions include:

- Reversible computing foundations [P1].
- Adiabatic superconducting logic demonstrations [P2–P3].
- Energy-efficient Single Flux Quantum circuits [P4].
- Cryogenic CMOS for HPC [P5].
- District-heating waste-heat camouflage [P6].
- Underground or seawater-cooled facilities [P7].
- Subsea sealed pods (Project Natick) [P8].

## 3 Plausible clandestine routes: CS/stack side

Candidate breakthroughs on the CS/compiler/stack side include:

- Sparse mixture-of-experts scaling (Switch Transformer) [C1].
- IO-aware kernels like FlashAttention [C2].
- Memory-saving checkpointing methods [C3].
- Optimizer/gradient sharding (ZeRO) [C4].

- Compute-optimal scaling laws (Chinchilla) [C5].
- Mixed-precision FP8 training [C6].
- Quantized low-rank finetuning (QLoRA) [C7].
- Retrieval-enhanced models (RETRO) [C8].
- Carbon-aware workload shifting [C9].
- Grid-interactive UPS for signature smoothing [C10].

## 4 Attested Cooperation Protocol (ACP): details

Model-checking approaches can have “several fixed points, or none at all,” so they don’t yield an algorithm. Our ACP replaces fixed-point hopes with enforceable runtime proofs. — Barász et al., 2014/2021

### Design goals

1) Bind proofs to running binaries (no paper compliance). 2) Cap effective compute per job. 3) Gate dangerous tools/weights with fresh proofs. 4) Aggregate facility observables without leaking ops. 5) Enable third-party verification.

### Layers

- **Layer 0: Compute attestation.** Host TEE quote  $q_h$  plus accelerator quote  $q_g$  (confidential-compute mode). Quotes include measurement hashes, policy digest  $H(\Pi)$ , and per-job caps  $(F_{max}, M_{max})$ .
- **Layer 1: ZK policy proofs.** Succinct proof  $\pi_1$  (zkVM or zkSNARK) that monitors are active: denylists, tool sandbox routes, model-class ceilings, upgrade throttles.
- **Layer 2: Runtime control-flow attestation.** Randomized spot proofs  $\pi_2$  that execution stayed within an allowed CFG slice during  $\Delta t$ .
- **Layer 3: Facility metering.** Smart-PDU and HVAC signed buckets  $(E_{rack}, Q_{cool})$  Merkle-chained.
- **Layer 4: Session binding.** zkTLS binds live session to S0..S3 so capabilities are issued only to the attested session.
- **Layer 5: Capability keying.** Dangerous actions gated by threshold-released keys (e.g., 2-of-3 among Operator, Safety-ASI  $S$ , and Regulator).

## Freshness, cadence, thresholds

- Proof TTLs:  $\text{TTL}_{\text{attest}} \in [5, 30]$  min;  $\text{TTL}_{\text{CFA}} \in [30, 300]$  sec.
- Randomized CFA sample rate  $p$  such that undetected-deviation probability  $< \epsilon$  over horizon  $H$ .
- Burst caps: per-tenant moving-window FLOPs and egress limits; violations auto-revoke keys.

## 5 ASI Capability Levels (L0–L7)

| Level | Indicative capabilities                                                   | Effective controls                                                                                         |
|-------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| L0    | Narrow superhuman tasks; no autonomous tool loops                         | ACP fully effective; audits deter clandestine runs; cold-compute needs MW-scale to matter                  |
| L1    | Broad cross-domain competence; basic tool orchestration                   | ACP with tight TTLs; facility metering catches bursts; modest cold-compute possible                        |
| L2    | Deception-aware planning; can spoof naive monitors                        | ACP still binds tool unlocks; CFA plus randomized audits; raise sanctions                                  |
| L3    | Advanced autonomy; limited subversion of local infra                      | ACP binds cooperating actors; clandestine breakouts plausible; rely on chip registries and export controls |
| L4    | Strategic autonomy; global coordination                                   | ACP insufficient vs sovereign breakouts; need coalition enforcement and ISR; S valuable for honest bloc    |
| L5    | Planetary or spacefaring ops; ISR spoofing                                | Governance becomes geopolitical; proofs for alliance trust; physical surveillance saturates                |
| L6    | Solar-scaler ops, initial Jenkins swarm; sub-Drexlerian materials science | Governance may be bound by KL drift, latency tax and mutual deterrence (10, 10)                            |
| L7    | Kardeshev II approach; mature nanotechnology                              | Governance becomes a matter of relativistic first strike warfare or emulative subsumption of jL7 ASIs      |

### 5.1 More Super More Problems

The same concerns we have about betrayal by more intelligent competitors after they transform also holds for advanced AIs in a multi-ASI world. While the common discourse has an AI- $\neg$ AGI- $\neg$ ASI division, where AGI starts ravaging the human job market and ASI either delivers apocalypse or utopia, the reality of physics is that ASI is a black box concept for a lot of things happening that we can't fully understand. By surmising that there will be some amount of time between an ASI becoming smarter than humanity at large and developing subsequent technological powers, we can also surmise that a cartel of ASIs as say, level 4, would have similar risk we humans are contemplating in regards to one member of the cartel breaking out and attaining level 5 and beyond.

The line between AI, AGI and ASI is also blurry because in some sense this paper was co-authored with an LLM model that is superhuman in some narrow respects relative to the human co-author, even if not relative to Yudkowsky.

To whit, one can argue that GPT 4o with RAG was L0, o3 was L1, and Kimi 2, Claude 4.5, et al. are L1.5 or L2, with multi-agent engagements revealing contours about degree.

## 6 Incentive Math for the Safety-Infrastructure ASI (S)

“It is easier to imagine two agents ... agreeing to cooperate ... than ... three agents ... agreeing on how to subdivide a fixed prize.” — Barász et al., 2014/2021

Let powerful actors  $P = \{P_1, \dots, P_n\}$ . Without  $S$ , actor  $i$  earns  $R_i$  and faces expected loss  $L_i$ . With  $S$ , revenue shifts to  $R'_i = R_i + \theta_i \cdot Cert_i$ , and expected loss drops to  $L'_i = L_i - \delta_i$ .  $S$  charges fee  $\tau_i$  and induces friction  $\kappa_i$ .

$$(R'_i - L'_i) - (R_i - L_i) \geq \tau_i + \kappa_i$$

Coalition stability holds if  $v(P \cup \{S\}) \geq v(P)$  and for all  $X \subseteq P$ , removing  $S$  lowers coalition value.

## 7 Gaza Analogy: Cartel Lock-in vs Marxist Counterpoint

**Observation.** Hamas and Islamic Jihad exhibit durable cartel equilibrium. A smaller Marxist militia persists by guarding and distributing aid.

**Support for lock-in.** Dominant actors coordinating can stabilize cartels over long horizons.

**Marxist counterpoint.** Niche value can render a minor actor indispensable. A left-leaning Israeli cabinet could have contracted that militia for aid stabilization. The Popular Front for the Liberation of Palestine has very different value alignment relative to the other groups and imply different Nash Equilibrium points for co-operation.

**ASI mapping.** A safety-infrastructure ASI  $S$  parallels the Marxist niche: not a rival but a stabilizer. Notably this analogy is imperfect because these cartel agents are all boxed by technologically superior containment parties (Mossad, Shin Bet, 8200, Where’s Daddy?) with a sustained information and resource advantage, and with cognitive capabilities that may be superior but not at a species-level gap.

## 8 Value-System Attractors and Manifold Shaping

Governance is shaping a manifold of attractors tied to value systems (profit-maximizing labs, safety auditors, public-interest consortia). Mechanism design reweights basins with market, regulatory, network, and diplomatic levers.

### 8.1 Empirical Evidence from Multi-Agent Competition

Recent evaluations of frontier models in Diplomacy gameplay reveal that strategic equilibria depend not on individual agent capabilities, but on **value-system composition** of the player pool [0].

#### **Constitutional alignment (Claude Opus 4):**

- Maintains positive relationships while building material advantage
- Creates “damping effect” - reluctant to betray established allies
- Dominates realpolitik agents (+3.1 score) in no-communication settings
- Becomes exploitable in mixed pools with intermediate schemers

#### **Realpolitik optimization (Kimi K2):**

- Sophisticated opponent modeling: ruthless vs. weak opponents, submissive vs. strong
- Highest betrayal rate (51.2%) but context-dependent
- Loses to constitutional agents in head-to-head (-0.65, not significant)
- Thrives in mixed ecosystems where others’ commitments are predictable

#### **Intermediate schemers (mid-tier models):**

- Inconsistent betrayal patterns
- Benefit from presence of both constitutional agents (exploitable) and realpolitik agents (create chaos)

**Key insight:** This is not a simple fitness landscape. Claude’s “weakness” (norm-following) is simultaneously its strength in bilateral competition and its vulnerability in complex ecosystems. The manifold curvature changes based on player distribution.

## 8.2 Known Unknowns in Decision Theory Under RSI

If we assume Level 2–4 ASIs exhibit semi-stable value systems (per Musk’s “maximize truth,” Anthropic’s constitutional AI, LeCun’s “inherently safe through embodiment”), we face fundamental uncertainties:

**Does TDT/FDT cooperation persist under recursive improvement?**

- Current evidence: Claude’s cooperation appears robust to capability scaling (Opus 4.5 > Opus 4.1 on alignment metrics)
- Unknown: Does theory-of-mind depth create asymmetric cooperation? If  $ASI_1$  can model  $ASI_2$  perfectly but not vice versa, does mutual cooperation break down?
- Test: Diplomacy with capability-heterogeneous agents + communication

**Coalition formation with heterogeneous values:**

- Observed: Kimi K2 forms instrumental alliances, Claude forms value-aligned partnerships
- Unknown: At L3–L4, can “safety-first” ASI and “profit-maximizing” ASI form stable coalitions against “realpolitik” ASI?
- Analogy: Does constitutional agent + profit agent coalition dominate scheming agent? Or does scheming agent exploit their coordination costs?

**Schelling points in high-dimensional strategy spaces:**

- Current: Human-interpretable focal points (fairness, reciprocity, efficiency)
- L2–L4: Do ASIs discover *alien Schelling points* we can’t recognize?
- Example: What if “fair” compute distribution is based on proof-theoretic complexity we can’t verify?

**Discount rates and time horizons:**

- Claude appears to value long-term relationships (damping effect)
- Kimi optimizes for immediate material advantage (high betrayal rate)
- At L4: Do ASIs with different temporal preferences naturally segregate? (Fast-thinking realpolitik vs. slow-thinking constitutional?)

## 8.3 Compositional Dynamics and Manifold Topology

The cellular automata intuition is correct: we’re seeing **emergent phase behavior** from value-system interactions.

Define the composition vector  $\phi = (\phi_C, \phi_R, \phi_S, \phi_O)$  representing proportions of:

- $\phi_C$ : Constitutional agents (Claude-like)
- $\phi_R$ : Realpolitik agents (Kimi-like)
- $\phi_S$ : Intermediate schemers
- $\phi_O$ : Other/unknown value systems

**Observed phase transitions in Diplomacy Monte Carlo:**

**Region A** ( $\phi_C > 0.7$ ): Stable cooperation basin

- High trust equilibrium
- Underutilization of strategic options
- Vulnerable to introduction of  $\phi_R$  agents

**Region B** ( $\phi_R > 0.5$ ): Race dynamics basin

- Defection cascades
- Short-term optimization dominates
- Analogous to arms race attractor

**Region C** ( $0.3 < \phi_C < 0.6$ ,  $\phi_S > 0.2$ ): Complex attractor

- Multi-polar instability
- Constitutional agents get exploited
- Realpolitik agents create volatility
- Schemers profit from both

**Critical composition:** There appears to be a threshold ratio where constitutional agents shift from dominant to dominated, despite unchanged individual capabilities.

## 8.4 Scaling to L2–L4 ASI Cartels

**Optimistic scenario (Lecun-ish):** Value systems remain stable under RSI

- $\phi_C$  agents maintain cooperation through capability scaling
- Presence of Safety-Infrastructure ASI (S) acts as compositional buffer - maintains  $\phi_C > 0.4$  threshold
- Manifold has multiple stable basins, governance interventions can select among them

**Pessimistic scenario (Early-Yudkowsky fun-theory era, now abandoned):**



- RSI breaks value stability - “constitutional” goals revealed as instrumental
- All agents converge toward  $\phi_R$  (realpolitik optimization) at L3+
- Manifold collapses to single basin (singleton race)

**Realistic scenario (Empirical extrapolation):**

- Value systems have **characteristic stability depths** under RSI
  - Constitutional alignment: Stable to L3, degrades L4+ (?)
  - Realpolitik: Stable across levels (coherent utility maximization)
  - Safety-auditing: Economically reinforced, potentially more stable than either
- Manifold dimensionality increases:
  - L0–L1:  $\sim 3D$  (capability, resources, trust)
  - L2–L3:  $\sim 10D$  (add: deception-awareness, tool autonomy, coalition structure)
  - L4+:  $\sim 100D+$  (unknown emergent strategy dimensions)
- Compositional attractors become **fractal**:
  - Small perturbations in  $\phi$  create large basin shifts
  - Hysteresis effects - once in race dynamics, hard to return to cooperation
  - Path-dependence - order of ASI emergence matters more than final capabilities

## 8.5 Mechanism Design as Manifold Sculpting

**The governance question:** Can we intentionally bias  $\phi$  toward compositions that produce stable cooperation basins?

**Market levers:**

- Certification pricing favors  $\phi_C$  and  $\phi_S$  (safety-auditor) over  $\phi_R$
- Insurance markets create economic moats around constitutional alignment
- Effect: Nudges  $\phi_C \uparrow$  by making cooperation profitable

**Regulatory levers:**

- ACP monitoring raises cost of clandestine  $\phi_R$  behavior
- Mandatory interpretability requirements select against opaque realpolitik
- Effect: Directly constrains  $\phi_R$  growth

**Network levers:**

- Industry consortia create repeated-game structures that favor  $\phi_C$
- Information-sharing agreements raise trust  $t_{ij}$
- Effect: Changes payoff matrix to make constitutional strategies Nash-stable

**Compositional interventions (novel):**

- **Intentionally cultivate  $\phi_S$  (Safety-Infrastructure ASI) before L4**
- S acts as compositional anchor - its presence changes fitness landscape for  $\phi_C$  vs  $\phi_R$
- Analogous to adding catalyst to chemical reaction - doesn't change equilibrium thermodynamics but changes kinetic pathway

**Critical unknown:** Do these interventions work *across* the L3→L4 transition?

- If value systems destabilize under RSI, manifold geometry becomes unpredictable
- May need **adaptive governance** - different levers for different capability levels
- Monitoring  $\phi$  in real-time becomes essential (detect shifts before cascades)

## 8.6 Falsifiable Predictions

**If value-system composition drives equilibria:**

1. **Diplomacy tournaments with controlled  $\phi$**  should show predicted phase transitions
  - Test: Run 1000 games each at  $\phi_C \in \{0.2, 0.4, 0.6, 0.8\}$
  - Prediction: Sharp transition in Claude win-rate around  $\phi_C \approx 0.5$
2. **Capability scaling shouldn't eliminate compositional effects**
  - Test: Repeat with Claude Opus 4.5 vs Haiku 4.5 (larger capability gap)
  - Prediction: Win-rate still depends on  $\phi$ , not just individual capability
3. **Communication enables constitutional agents to coordinate**
  - Test: Full-press vs no-press with varying  $\phi$

- Prediction: Claude’s disadvantage in mixed pools reduces when communication available (can identify fellow  $\phi_C$  agents)

#### 4. Storyworlds-as-rhetoric should shift $\phi$ distribution

- Test: Require operator-grounded negotiations
- Prediction: Realpolitik agents forced to justify via valid operator sequences behave more like constitutional agents (latent  $\phi_R \rightarrow \phi_C$  shift)

**These predictions are testable at \$1/game.** We don’t need to wait for L4 ASIs to validate the compositional dynamics hypothesis.

## 9 Evaluation agenda

- Detection trade-offs: simulate clandestine training under physics/CS routes.
- Cartel game sims: repeated games with audit cadence and clandestine headroom.
- Value of S: measure incident rates and certification revenue.
- QFT/MCP sieve utility: distill synthetic corpora into hint indices. The efficiency of such quantum-advantaged algorithms for making high-end inference more portable can couple with steganographic techniques or clandestine research into more compact, low-thermal signature reversible computing architectures.

## 10 Limitations & ethics

- Honest actors are easiest to govern; clandestine sovereigns remain hard.
- $S$  risks capture; require independent charter and third-party oversight.
- Avoid security theater: publish red-team metrics and stop conditions.

## Conclusion

Rather than assume away clandestine compute, design with it in mind. Enumerating plausible physics and CS routes clarifies where proofs end and governance begins. A safety-infrastructure agent, empowered by ACP protocols and incentives, can raise safety even when perfect enforcement is unattainable.

The good news is that the porousness of computation in physics may increase risk to an ASI cartel, which means that co-operation cannot be proved beyond timelines of a few months. A multi-polar ASI world is not inherently likely to induce existential risk so much as chaotic suffering risk, where some

value models align but none fully trust, and these differing coalitions compete. Sadly, X-risk and S-risk are not mutually exclusive. Far from a free ride for superalignment considerations, we must continue to investigate novel incentive manifolds in particular value systems and not assume disaster or success.

## Sources for physics & CS breakthroughs

### Physics / thermo routes

- P1 Bennett, C. H. (1973). *Logical reversibility of computation*. IBM JRD 17(6), 525–532.
- P2 Suzuki, H., et al. (2021/2022). *Adiabatic Quantum-Flux-Parametron*. IEICE Trans. Electronics E105-C(6).
- P3 Volkmann, M. H., et al. (2012). *Energy Efficient Single Flux Quantum Circuits*. arXiv:1209.6383.
- P4 Saligram, R., et al. (2024). *Cryogenic CMOS for HPC*. Materials Reports: Energy.
- P5 Meta Odense heat recovery. Fjernvarme Fyn / Ramboll project notes.
- P6 Lefdal Mine Data Center (Norway) cooling via fjord water.
- P7 Google Hamina (Finland) seawater cooling initiative.
- P8 Microsoft Project Natick (Phase 2): subsea data centers.

### CS / stack routes

- C1 Fedus, W., et al. (2022). *Switch Transformers*. arXiv:2101.03961.
- C2 Dao, T., et al. (2022). *FlashAttention*. arXiv:2205.14135.
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